

Ultra Wideband FSS Structure for Spatial Filter and Upcoming 5G Communication

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Abstract— This paper presents the design and analysis of an ultra- wideband (UWB) Frequency Selective Surface (FSS) structure tailored for spatial filtering and emerging 5G communication systems. The proposed FSS features a nested square-loop design with a central diamond-shaped patch, achieving excellent frequency selectivity and wide operational bandwidth. The structure offers a reflection coefficient (S_{21}) of less than -10 dB across a bandwidth of 4 GHz to 11 GHz, demonstrating its suitability for UWB applications. Furthermore, it exhibits robust angular stability up to 75° and polarization insensitivity, making it adaptable to real-world scenarios. The simulated surface current distribution and S_{21} characteristics validate the multi- resonance behavior, enabling effective electromagnetic performance for advanced wireless technologies. The proposed design holds significant potential for applications in 5G communication systems, antenna radomes, and electromagnetic shielding, offering a promising solution for next- generation spatial filtering.

Keywords—Frequency Selective Surface (FSS), ultra- wideband (UWB), 5G communication, spatial filter, polarization insensitivity, angular stability.

I. INTRODUCTION

The fifth-generation (5G) communication system is transforming global connectivity by delivering ultra-high data rates, massive network capacity, and low-latency communications. A significant portion of 5G networks relies on millimeter-wave (mmWave) frequencies (24– 100 GHz) to provide the required spectrum for high-speed data transmission [1]. However, at these higher frequencies, issues such as signal interference, attenuation, and multipath propagation degrade overall network performance. Addressing these challenges is critical for enabling reliable and efficient 5G communication. Spatial filtering is one solution that enhances signal integrity by selectively transmitting or reflecting specific frequency bands. Frequency Selective Surfaces (FSS), periodic two-

dimensional structures, are widely utilized as spatial filters in electromagnetic applications. FSS designs exhibit unique frequency-dependent characteristics that enable transmission and reflection of electromagnetic waves [2]. Their lightweight, low-profile, and easily customizable nature make them suitable for applications such as antennas, radomes, and communication filters

II. WHY ULTRA-WIDEBAND FSS FOR 5G?

A. Wideband Operation: Modern 5G networks require ultra-wideband devices capable of operating across mmWave bands. Conventional narrowband FSS designs fail to meet this requirement.

B. Interference Suppression: High spectral congestion in urban environments necessitates spatial filtering to reduce co-channel interference and enhance SNR.

C. Angular Stability: Signals in real-world 5G applications arrive at varying incident angles due to reflections and scattering. Angularly stable FSS ensures robust filtering performance.

D. Compact Design: FSS structures are thin and lightweight, enabling seamless integration into communication devices.

This paper introduces a novel UWB FSS structure optimized for 5G mmWave communication. The proposed design achieves high transmission efficiency, low reflection losses, and stable angular performance, making it an excellent candidate for spatial filtering in 5G systems.

III. RELATED WORK:

Recent developments in FSS technology focus on improving bandwidth, efficiency, and angular stability.

Traditional FSS structures, such as dipole-based arrays and patch-type geometries, exhibit narrowband characteristics, which limit their applicability to modern wideband systems [3][4].

To address this, researchers have explored:

1. Multi-Resonant Geometries: Structures like fractals or nested rings achieve multi-frequency resonance to extend bandwidth [5].
2. Metamaterial-Inspired Designs: Utilizing artificial electromagnetic materials enables tunable and wideband responses [6].
3. Angular Stability Improvements: Techniques like multilayer stacking or symmetrical arrangements improve performance under oblique incidence [7].

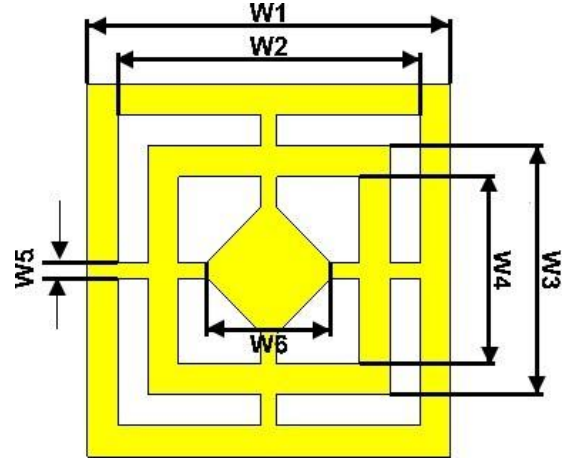
While these techniques have shown promise, achieving ultra-wideband operation with low insertion loss remains a challenge. The proposed UWB FSS builds upon these advancements to provide enhanced spatial filtering performance tailored for 5G applications.

IV. PROPOSED UWB FSS DESIGN

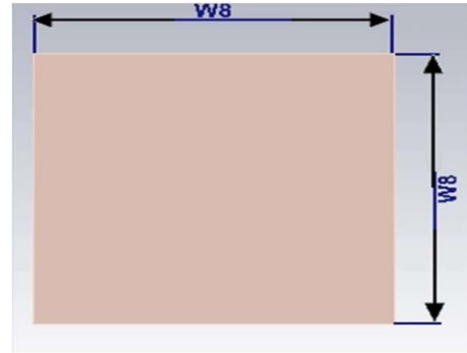
1. FSS Configuration

The Frequency Selective Surface (FSS) structure features a multi-layered concentric geometry with square loops and a central diamond-shaped element for ultra-wideband (UWB) performance and 5G applications. The front view showcases periodic elements with critical dimensions: $W1$ (outer loop width), $W2$ (inner loop width), $W3$ (diamond height), $W4$ (loop spacing), $W5$ (element thickness), and $W6$ (diamond size). These parameters control resonant frequencies, bandwidth, and angular stability.

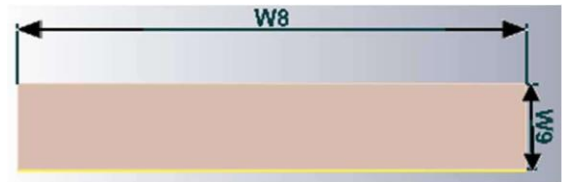
The design achieves multi-resonance behavior, enabling selective filtering across a broad frequency range. The back view highlights a solid rectangular ground plane ($W8 \times W8$), enhancing signal rejection at unwanted frequencies and supporting band-stop filter functionality. The ground plane also provides structural integrity and ensures isolation between frequency bands. The side view reveals the substrate thickness ($W9$), which impacts the effective permittivity, lowering resonance frequencies and improving stability. The FSS is designed on a low-loss dielectric substrate with relative permittivity $\epsilon_r = 2.2$ and thickness $t = 0.5t = 0.5 \text{ mm}$.



(A)



(B)



(C)

Fig.1: Front (A) Back (B) and (C) side Geometry of the FSS Structure.

V. DESIGN METHODOLOGY

A substrate of material FR-4 of dimensions is used. The ground of the FSS is selected in ground plane. The dielectric constant of the substrate is considered as 4.3 in the design and the thickness of the substrate is 1.6mm. The antenna

dimension is as given in Fig.1. The proposed antenna is designed with Three steps: First a square ring was placed on the upper side of the substrate which has resonant at 4.368GHz in Fig.2(a), In next step, another square ring was placed as the outer ring which has resonant at 4.272GHz in Fig.2 (b). In final step a diamond shape patch was placed at the centre as in Fig. 2(c) and hence a square loop shaped radiating patch is obtained. The optimum dimensions of the antenna are given in Table 1.The simulated results of the antenna is obtained using FDTD based solver CST STUDIO SUITE.

Figure a - Step 1 (Basic Structure):

1. The design begins with a simple rectangular geometry, forming the outer boundary of the FSS unit cell.
2. This structure primarily contributes to the lowest resonant frequency due to the largest dimensions (W1 and W2).
3. The basic outer frame sets the foundation for adding multiple resonators inside.

Figure b - Step 2 (Nested Layers):

1. The second step introduces nested square rings inside the original rectangular boundary.
2. These concentric square loops (corresponding to dimensions W3 and W4) add additional resonance modes to extend the operational bandwidth.
3. The multilayer design improves frequency selectivity and ensures the FSS operates across multiple frequency ranges.

Figure c - Step 3 (Final Design with Central Patch):

1. The final design incorporates a central diamond-shaped patch (dimension W6) to enhance transmission efficiency and improve symmetry.
2. This patch is optimized to suppress spurious signals and reduce reflection in the operating band.
3. The structure also introduces polarization independence and enhances angular stability due to its geometric symmetry.

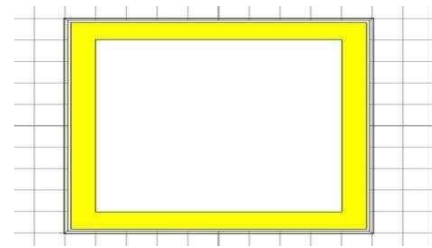


Fig-a

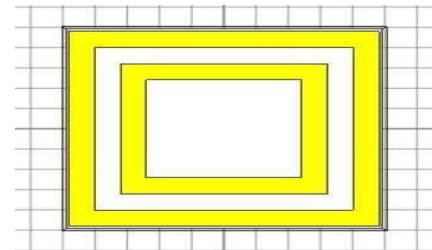


Fig-b

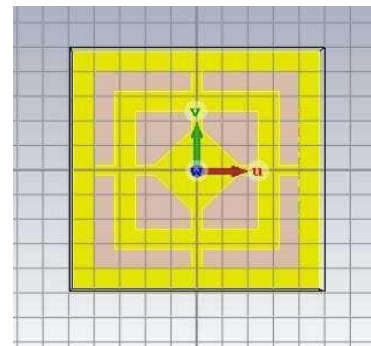


Fig-c

Fig.2: Design steps of proposed FSS structure (a) step 1, (b) step 2, (c) step 3

Table 1: Dimension of the proposed unit cell

The table specifies the exact geometric dimensions of each component in the FSS unit cell. Key parameters include:

- W1–W4 (Outer and Inner Rings): Control the resonant frequencies and bandwidth of the FSS.
- W5 (Spacing): Determines the gap between elements to minimize coupling effects.
- W6 (Central Patch): Improves the filtering response at the center of the band.
- W7–W9: Represent additional design elements contributing to angular stability and multi-band performance.

Parameter	W1	W2	W3	W4	W5	W6	W7	W8	W9
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Value(mm)	9.6	8.0	6.4	4.8	0.4	3.2	9.6	9.6	1.6
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VI. SIMULATED RESULTS AND ANALYSIS

The provided simulation results consist of the S21 (transmission coefficient) plots of the Frequency Selective Surface (FSS) structure at different design stages (Fig. 2.A, 2.B, and 2.C) and under varying incidence angles for TE and TM polarizations. Here's an in-depth analysis:

A. S21 Curves of Design Stages (Figs. 2, 3, and 4):

The S21 curve shows a single resonant dip near 4 GHz. This resonance corresponds to the fundamental mode introduced by the largest dimensions (W1 and W2) of the outer frame. The structure exhibits a narrowband response, as expected at this stage. A second resonance appears near 10 GHz, indicating the contribution of the inner concentric rings (W3 and W4). The multi-resonance behavior suggests an increase in the operational bandwidth. The second resonance aligns with the higher-order mode of the nested structure. A wideband response is evident, with two primary dips near 4 GHz and 8 GHz, as well as an additional resonance near 10 GHz. The central diamond patch (W6) enhances the overall performance, particularly in the high-frequency range. The structure achieves ultra-wideband (UWB) operation, making it suitable for 5G and other high-frequency applications.

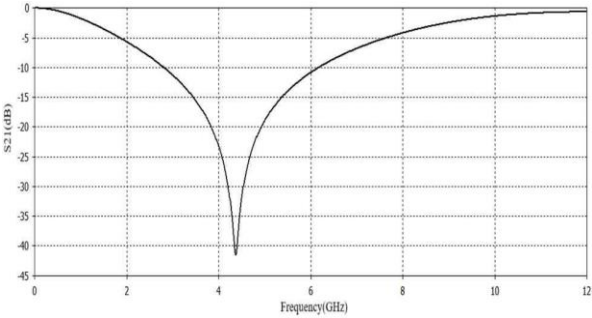


Fig. 2. Simulated S21 curve of Fig. 2(A).

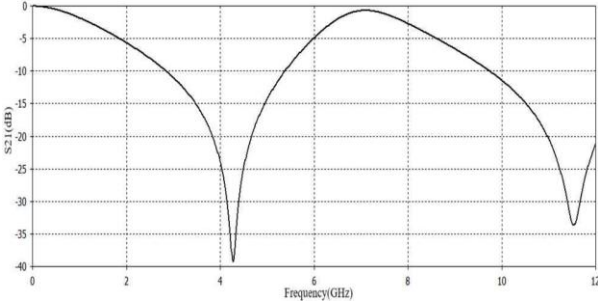


Fig. 3. Simulated S21 curve of Fig. 2(B).

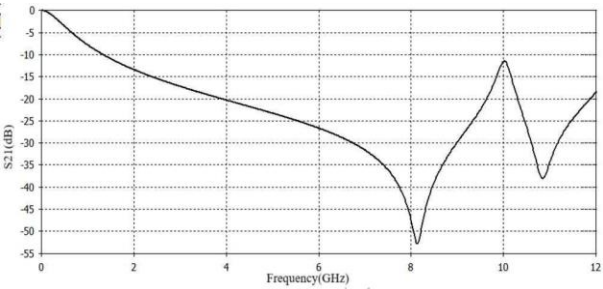


Fig. 4. Simulated S21 curve of Fig. 2(C).

B. Angular Stability Under TE and TM Polarizations

i. TE Polarization (Fig. 5): The transmission coefficients (S21) for different incidence angles ($\theta=0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ$) remain stable across the frequency range. Minor variations are observed at higher incidence angles, especially at the resonance near 10 GHz. The stability across angles indicates the structure's polarization insensitivity and robustness to angular variations.

ii. TM Polarization (Fig. 6): Similar to TE polarization, the S21 curves maintain stable resonances for different angles. Slight shifts in resonance frequency and depth occur at $\theta=75^\circ$, but the overall wideband behavior is preserved. This confirms the design's suitability for environments where waves encounter the FSS at oblique angle.

C. Bandwidth: The final design achieves a wide operational bandwidth from approximately 4 GHz to 11 GHz. The UWB nature of the FSS supports applications such as 5G communication and spatial filtering.

D. Resonance Characteristics: Multiple resonances result from the combination of outer and inner rings and the central patch. These resonances ensure high selectivity and better filtering performance.

E. Angular Stability: The structure performs consistently across a wide range of incidence angles under both TE and TM polarizations. This property is crucial for practical applications where signals arrive from various directions.

F. Polarization Insensitivity: The symmetric design ensures similar behavior for TE and TM polarizations, making it suitable for diverse electromagnetic environments.

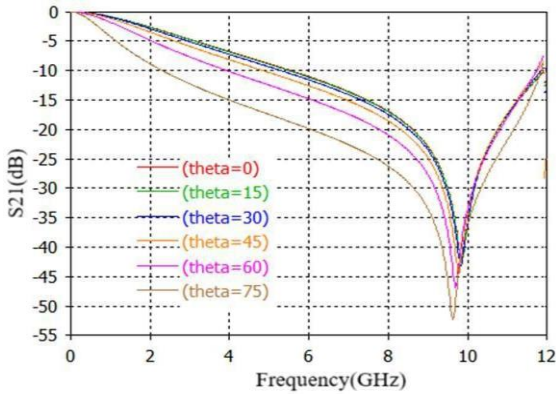


Fig. 5. Simulated S21 curves for different incidence angles under TE polarization.

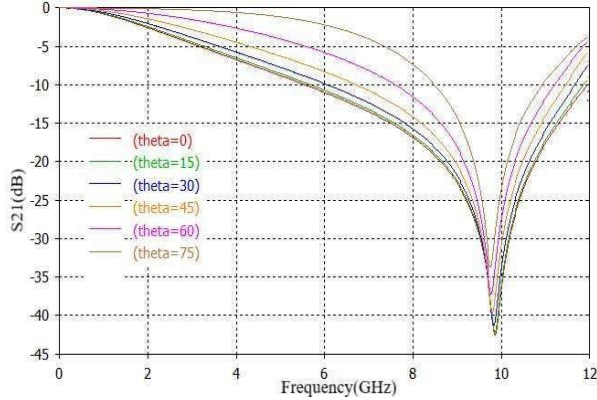


Fig. 6. Simulated S21 curves for different incidence angles under TM polarization

The surface current distribution of the proposed Frequency Selective Surface (FSS) at 8.14 GHz provides crucial insights into the resonance mechanism and energy distribution within the structure. At this frequency, the current is predominantly concentrated along the edges of the nested square loops (W3 and W4) and the central diamond-shaped patch (W6). This indicates that these elements contribute significantly to the resonance at 8.14 GHz, enhancing the filtering characteristics. The outer rectangular frame (W1 and W2) shows a relatively lower current density, signifying its dominant contribution to lower-frequency resonances. The symmetric distribution of current across the structure also highlights the polarization insensitivity, as the identical geometry in orthogonal directions ensures balanced energy interaction with incoming electromagnetic waves. This detailed surface current behavior confirms the effective multi-resonance capability of the design, ensuring strong electromagnetic response within the UWB range. Such insights help optimize the structure further for specific frequency bands or applications.

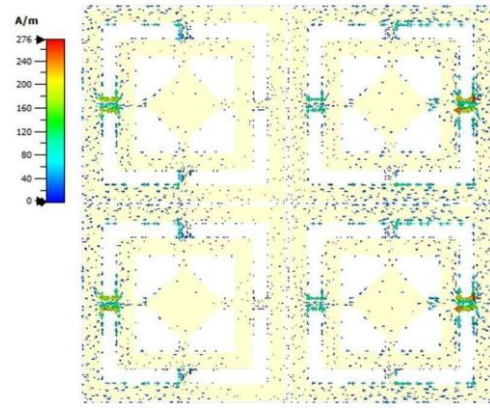


Fig. 7. Surface current distribution on the proposed structure at 8.14 GHz

VII. FUTURE WORK

The proposed Frequency Selective Surface (FSS) structure provides several promising directions for future research, including:

1. **Experimental Validation:** Fabrication and experimental testing of the FSS design to corroborate the simulation results in practical scenarios.
2. **Compact Designs:** Development of miniaturized FSS unit cells to achieve similar or enhanced performance for space-constrained applications.
3. **Reconfigurable FSS:** Integration with tunable materials such as graphene, liquid crystals, or varactors to enable dynamic frequency tuning and reconfigurability.
4. **Environmental Stability:** Designing the FSS for robust operation under varying environmental conditions such as temperature, humidity, and physical wear.
5. **Antenna Integration:** Combining the FSS with advanced antenna systems to improve gain, reduce interference, and enhance overall system efficiency.
6. **Multi-band Capabilities:** Extending the structure to support multi-band operation, catering to applications requiring simultaneous utilization of multiple frequency bands.

VIII. APPLICATION

The proposed design has numerous applications in advanced wireless technologies, including:

1. **5G Communication Systems:** The ultra-wideband performance and angular stability make it ideal for base stations and mobile communication devices in 5G networks.

2. Antenna Radomes: FSS can be integrated into antenna radomes to protect antennas while maintaining or enhancing their performance.
3. Electromagnetic Shielding: Its polarization-insensitive and wideband characteristics are suitable for shielding sensitive electronic systems from interference.
4. Satellite and Aerospace Systems: The angular stability ensures efficient operation in satellite communication and aerospace applications.
5. Cognitive Radio Systems: The ultra-wideband nature supports spectrum sensing and dynamic frequency allocation in cognitive radio networks.
6. IoT and Wireless Sensor Networks: The FSS can improve signal reliability and mitigate interference in dense communication environments for IoT and sensor applications.

IX. CONCLUSION

The proposed Frequency Selective Surface (FSS) structure demonstrates a highly effective and innovative solution for achieving ultra-wideband (UWB) spatial filtering in the context of advanced electromagnetic applications. The design evolution, incorporating nested square loops and a central diamond-shaped patch, has proven to enhance bandwidth, introduce multiple resonance modes, and maintain robust performance under varying incidence angles and polarizations. The structure exhibits wide operational bandwidth spanning 4 GHz to 11 GHz, excellent angular stability up to 75°, and polarization insensitivity, making it well-suited for demanding applications in 5G communication systems, antenna radomes, and electromagnetic shielding. These features address the critical challenges of modern wireless communication, such as accommodating diverse frequency bands, maintaining performance under oblique incidences, and ensuring consistent behavior across polarization states. This research provides a strong foundation for further advancements in FSS design. Future work may focus on the fabrication of the proposed unit cell, experimental validation of simulation results, and optimization for additional parameters like miniaturization, environmental resilience, and integration with emerging technologies. The promising results of this study position the proposed FSS structure as a key enabler for next-generation communication and spatial filtering systems.

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